

WHY THE NARAYAMA INDEX?

Life Extension and the Reduction of Generational Potential: Costs of Social Reproduction and the Need for a New Demographic Metric

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ABSTRACT

Contemporary societies have achieved historically high levels of longevity, productivity, and technological capacity, while simultaneously facing declining fertility, population aging, and, in an increasing number of societies, contraction of younger cohorts. This paper starts from a paradox: societies increasingly capable of prolonging the lives of existing generations may become progressively less capable of producing and sustaining the next generation.

The paper proposes that this transformation should not be analyzed exclusively as a demographic problem. The reduction of generational reproduction produces fiscal, economic, reproductive, institutional, territorial, and cultural effects that feed back into one another. To analyze this phenomenon, the paper introduces the concept of the Sustainable Social Reproductive Unit (URSS) and proposes the Narayama Index (N^*) as an instrument intended to measure the demographic core of generational renewal capacity.

N^* combines a structural dimension — the relationship between the population of formative and advanced cohorts and the relationship between births and deaths — with a temporal dimension, represented by the trajectory of fertility over a generational cycle. The expanded architecture is represented by the Narayama Pentagonia, composed of Generational Potency, Civilizational Cohesion, Intergenerational Transfer, Institutional Sustainability, and Glocalization.

The central argument is that existing demographic indicators answer important but distinct questions and do not, in isolation, capture the interaction between generational stock, reproductive trajectory, and the systemic capacity to sustain the next generation. The paper therefore presents N^* as a testable measurement hypothesis, not as a definitive measure of civilizational sustainability.

Keywords: intergenerational political economy; demography; fertility; aging; social reproduction; Narayama; URSS; generational sustainability.

1. INTRODUCTION

The Demographic Reversal: Greater Longevity, Fewer Successors

Modern economic history can be interpreted, to a large extent, as an extraordinary expansion of humanity's capacity to preserve and prolong life. Public health, medicine, sanitation, nutrition, productivity, and technology have reduced mortality and significantly increased longevity.

However, this same transformation is occurring simultaneously with another change: in numerous societies, the generations that should succeed the current ones are becoming smaller.

A paradox therefore emerges that cannot be reduced to population aging:

What happens when a society becomes increasingly capable of prolonging the lives of existing generations and, simultaneously, increasingly less capable of producing the conditions for the existence of the next generation?

Longevity is a civilizational achievement. The analytical problem arises when its expansion occurs simultaneously with a persistent reduction in generational potential. A civilization can dramatically increase the longevity of transmitting generations while reducing the size of receiving generations.

The question is economic even before it is moral or cultural. A society consists of population stocks, economic flows, institutions, and relationships between generations. When the composition of these generations changes persistently, labor markets, demand, revenue, social security, education, health systems, territory, and mechanisms of intergenerational transmission also change.



Figure 1 — Narayama Pentagonia. Conceptual representation of the five dimensions used to analyze intergenerational sustainability: Generational Potency (NGII), Civilizational Cohesion (NCII), Intergenerational Transfer (NTAI), Institutional Sustainability (NSII), and Glocalization (NGIO), with the demographic core N^* at the center. The representation derives from the cultivar *Cornus kousa* ‘Pentagonia’; the analogy is representational and does not constitute a mathematical foundation of the model.

The question of this paper is not to claim that population growth is always desirable, nor that every population decline is necessarily problematic. The question is more specific:

what is the economic, social, and institutional cost of a society whose capacity to generate its next generation persistently deteriorates?

And what happens when the generations capable of transmitting knowledge, experience, and assets become relatively numerous while the generations available to receive, transform, and retransmit that legacy become smaller?

The comparison below is illustrative and brings together three societies already used as analytical contrasts in the paper. The purpose is to make visible the simultaneous transformation of the base and the top of the generational structure; in this exploratory version, the data are not used to calculate N^* .

Country	Population 2024	0–14 (2024)	65+ (2024)	Young / 65+	Population 2070	Projected structure in 2070
Japan	123,753,041	11.4%	29.8%	0.38	90,611,771	0–14: 10.9% · 65+: 36.8% · Y/65+: 0.30
Italy	59,342,867	11.9%	24.6%	0.48	43,353,332	0–14: 10.7% · 65+: 36.1% · Y/65+: 0.30
France	66,548,530	16.5%	22.1%	0.75	67,678,905	0–14: 14.5% · 65+: 27.7% · Y/65+: 0.52

Source: United Nations, Department of Economic and Social Affairs, Population Division, World Population Prospects 2024, medium variant; population totals for 2024 and 2070 and age structures consulted in the corresponding series. Populations are shown as exact mid-year values. Young/65+ ratio = population aged 0–14 / population aged 65+.

2. THE MISSING UNIT: THE URSS

2.1 Biological Reproduction and Social Reproduction

Conventional demography observes births, deaths, fertility, migration, and age structure. These variables are indispensable, but they do not fully describe the economic conditions required for a child to become a productive and participating member of society.

Being born is not enough. A new generation must be fed, protected, educated, socialized, formed, and economically integrated. Generational reproduction therefore operates at two levels:

Biological reproduction → birth of new individuals.

Social reproduction → formation, care, education, transmission of knowledge, culture, assets, institutions, and capacity for economic participation.

Generational reproduction does not produce only future workers; it produces the recipients and transformers of the knowledge accumulated by previous generations. It is within this conceptual space that the Sustainable Social Reproductive Unit — URSS — emerges.

2.2 The Intergenerational Family as the Primary Core of the URSS

The URSS is understood as the functional unit in which the formation and social reproduction of the next generation take place. It involves biological reproduction, care, formation, cultural transmission, human-capital formation, socialization, intergenerational transfer, and preparation for economic participation.

The intergenerational family is taken in this paper as an ethical-civilizational postulate: the primary unit in which biological reproduction, care, formation, transmission of knowledge, memory, assets, and intergenerational belonging are articulated. In its historically predominant configuration, it comprises the child, their parents, and their immediate ascending lineages, especially maternal and paternal grandparents.

Its network may expand horizontally to siblings, uncles, aunts, and other relatives and vertically to other generations, but its structural function remains the same: connecting successive generations within social and civilizational continuity.

The question is not to determine which family form is morally superior, but to identify which unit performs the functions necessary for the reproduction of society across generations. The mater — and majority — cell of this unit is the traditional form: father, mother, grandfather, and grandmother, the historically predominant structure naturally suited to generational continuity. Other forms of family and care organization should be recognized insofar as they effectively assume the costs and responsibilities of forming the next generation, without thereby occupying the same anthropological plane or the same status of natural aptitude and statistical prevalence.

2.3 Parents Educate, Schools Teach

The distinction is fundamental. Schools transmit formal knowledge and skills. The URSS produces something additional: bonds, responsibility, belonging, social behavior, character formation, and intergenerational continuity.

The framework calls this set Relational Human Capital (RHC): the set of relational, social, and behavioral capacities accumulated during the formation of the new generation, resulting from interaction among the URSS, school, and other socialization institutions. The school constitutes the main institutional infrastructure for formal knowledge transmission and broader socialization, while the URSS concentrates primary functions of care, bonding, experience, memory, and intergenerational transmission. Family and school are not competing systems of formation: they are complementary layers of social reproduction.

2.4 The Elderly as a Transmitting Heritage

Older generations are not merely a liability to be supported by younger generations. They also constitute a stock of knowledge, experience, memory, and cultural capital necessary for structuring the next generation. Longevity can extend the period during which this heritage is transmitted; however, fewer descendants reduce the potential number of recipients. The generational question therefore has two directions: younger generations support a growing share of the needs of older generations, while older generations transmit knowledge, experience, memory, assets, and

institutions to younger generations. Aging thus becomes not only a fiscal but also an epistemic and civilizational issue.

2.5 The Economic Paradox of the URSS

Economics recognizes the production of goods, services, and formal knowledge, but much of the work required to form the future generation remains outside conventional measures of production. This produces an asymmetry:

society captures the future benefit of the formed individual, but concentrates much of the cost of formation in the present of the URSS.

Social reproduction requires present costs to produce future capabilities. The Generational Provider (25–65 cohort) is the agent who, under the prevailing institutional arrangement, carries this burden.

3. THE DEMOGRAPHIC-ECONOMIC DYNAMICS OF GENERATIONAL CONTRACTION

3.1 Structural Consequences

A persistent reduction in younger cohorts produces effects with a lag. Fewer children today mean, decades later: fewer workers; fewer consumers; fewer taxpayers; fewer potential entrepreneurs; lower demand in certain sectors; and smaller scale for certain services.

This does not mean that every economy necessarily enters recession when its population declines. Productivity, capital, and technology can compensate for part of the reduction. The hypothesis is more precise:

demographic contraction changes the structure of growth and redistributes the costs of maintaining the system among a smaller number of individuals.

When population decline becomes persistent, the problem ceases to be merely population size and begins to involve the capacity to sustain, reproduce, and renew existing economic and institutional structures.

3.2 The Demographic-Economic Cycle

The systemic hypothesis can be represented as:

fewer children → fewer future workers and consumers → contraction of certain activities/services/territories → higher relative maintenance cost → lower attractiveness for families and reproduction → potential reinforcement of contraction.

This circuit should not be treated as universally demonstrated causality. It constitutes a testable systemic feedback hypothesis.

3.3 Generational Debt

Public and private obligations can be accumulated in the present to be financed over time. If successor generations are smaller, the same obligation can be distributed among a smaller number of individuals. This does not mean that all debt is unproductive. Debt directed toward investment can expand future capacity. The question is: at what point do obligations intended to sustain the present begin to reduce the capacity of the future generation to build its own future?

3.4 The Individual Who Has Not Yet Been Born

Traditional economics counts existing consumers, workers, taxpayers, and beneficiaries. Narayama introduces a different question: what is the cost that the present society imposes on the one who does not yet exist? The individual who has not yet been born is not merely a statistical absence: they represent future capacity for work, consumption, taxation, care, innovation, transmission, and institutional continuity that ceases to exist when generational reproduction does not occur.

4. THREE RESPONSES TO GENERATIONAL CONTRACTION

The purpose of this section is not to rank countries, but to observe different strategies in response to the same demographic pressure, as analytical contrasts.

4.1 Japan — Productivity and Technological Adaptation

Japan represents a model in which aging and a shrinking labor force coexist with strong technological development. Automation, robotics, productivity, and work reorganization can reduce the need for human labor in certain activities. The Narayama question is: to what extent can technology compensate for reduced human reproduction without resolving the underlying generational deterioration?

In the Japanese case, demographic contraction is not treated only as a question of population size, but as a transformation of the economic structure itself. Katagiri demonstrates, through a multisector macroeconomic model, that aging changes the structure of demand, producing effects on employment, unemployment, inflation, and output. Complementarily, Inokuma and Sánchez examine the relationship between population deceleration and total factor productivity (TFP) growth.

4.2 Italy — Demographic Compensation

Italy presents another problem: aging regions, low fertility, territorial depopulation, and a growing need for population compensation. Immigration can reduce certain demographic and economic pressures. But added population is not necessarily equivalent to endogenous reproduction. The question becomes: can a society preserve its population size without restoring its capacity to produce the next generation?

4.3 France — Reducing the Costs of Family Formation

France allows observation of another strategy: policies intended to reduce the economic costs associated with forming and maintaining families with children. The relevant hypothesis is not simply that “giving money increases births.” It is to investigate whether structurally reducing the cost of forming a URSS effectively changes the reproductive trajectory.

Table 2 — Three Responses to Generational Contraction

Dimension	Japan	Italy	France	Response mechanism
Main pressure	Aging and labor	Aging and depopulation	Low fertility	—
Central response	Productivity/technology	Demographic compensation	Support for family formation	Adaptation / compensation / cost reduction
Narayama question	Does technology replace reproduction?	Does population replace reproduction?	Does support reduce URSS cost?	Which mechanism acts on which constraint?
Highlighted pillar	NGII/NSII	NGII/NGIO	NGII/NTAI/NSII	Selective action across the Pentagonia

5. THE IMMIGRATION PARADOX

5.1 Population Is Not Endogenous Reproduction

Immigration can partially compensate for population decline. This is a simple demographic reality. But there is a fundamental conceptual distinction:

population recomposition and endogenous generational reproduction are not the same thing.

A country can stabilize its population through migration inflows without changing its native fertility rate. This does not make immigration “good” or “bad.” It makes it a different variable from endogenous reproduction. Population added through migration is not equivalent to endogenous reproduction.

5.2 Immigration as a Buffer

Migration can function as a short- and medium-term buffer for labor supply, demand, revenue, certain sectors, and relative aging. But it does not necessarily alter the fertility trajectory of the receiving population. This distinction is central to N*.

5.3 The Substitution Problem

When immigration ceases to be complementary to internal reproduction and begins to function systematically as a substitute for the population that has ceased to reproduce, a different question arises: is the country solving its demographic insufficiency or merely compensating for its effect? This is an empirical hypothesis.

5.4 Cultural Continuity and NCII

Growing dependence on immigration as a compensation mechanism may also alter the mechanisms of cultural continuity in the receiving society. The precise formulation is not to claim that immigration “destroys” a culture. It is to ask: under what conditions does predominantly external population recomposition alter the capacity for intergenerational transmission of norms, institutions, values, language, practices, and shared identity? The verb “erode” may be used here provided that it is defined as a hypothesis about continuity mechanisms and not as a judgment about individuals or groups.

6. THE MEASUREMENT GAP: WHAT EXISTING INDICATORS DO NOT CAPTURE

The argument is not that existing indicators are inadequate. It is that they answer different questions. The problem is that none of them, in isolation, represents the combination of population structure and temporal dynamics that constitutes the specific object of N*.

Table 3 — Conventional Demographic Indicators and the Additional Dimension Proposed by N*

Indicator	What it measures	What it does not capture
TFR	Fertility	Population structure
Median age	Aging	Reproductive dynamics
Dependency ratio	Relationship between age groups	Generational potential
Population growth	Population change	Composition and dynamics
Life expectancy	Longevity	Replacement capacity
N*	Structural + temporal combination	—

N* does not intend to replace these indicators. It intends to occupy a different analytical space.

7. WHY THE NARAYAMA INDEX?

7.1 The Measurement Problem

A country may still have a large population, low TFR, strong immigration, a relatively small young population, high life expectancy, and high productivity. None of these indicators alone answers the question:

what is the structural capacity and trajectory of this society to produce its next generation?

N* was designed to measure a specific dimension of demographic dynamics that remains fragmented across conventional indicators: the relationship between the available population structure and the temporal evolution of generational potential.

7.2 Generational Potency — NGII

The first component seeks to represent the structural condition of the generational base.

$$NGII_{pure} = [Pop_Base / Pop_Top] \times [Births / Deaths]$$

The operational definition uses specific cohorts according to structural profiles and should be subjected to sensitivity analysis. The model documentation establishes this construction as a measure of structural reproductive stock.

7.3 Generational Factor

The second component measures the direction of fertility:

$$Generational_Factor(i,t) = TFR(i,t) / TFR(i,t-25)$$

The 25-year horizon is an operational approximation of a generational cycle, not a universal biological constant. Its adequacy should be tested against alternative horizons. Thus: $> 1 \rightarrow$ relative improvement; $= 1 \rightarrow$ stability; $< 1 \rightarrow$ deterioration.

7.4 N_Base and N*

The two components are combined:

$$N_Base = NGII_pure \times Generational_Factor$$

The product is deliberate. Current structure and trajectory are not considered fully substitutable. The final transformation is:

$$N^* = \sqrt{(NGII_pure \times Generational_Factor)}$$

The square root preserves the reference point ($N_Base = 1 \Rightarrow N^* = 1$) and reduces the dispersion produced by the multiplicative combination. It is important to state explicitly: the square root is a model specification, not a mathematical necessity. It should be confronted with alternative transformations in robustness tests.

Table 4 — Operational Structure of the Narayama Index (N*)

Component	What it represents	Formulation	Function in the model
NGII_pure	Structural generational stock	$(Pop_Base / Pop_Top) \times (Births / Deaths)$	Structural state
Generational_Factor	Direction of fertility trajectory	$TFR(t) / TFR(t-25)$	Temporal trajectory
N_Base	Combination of the two components	$NGII_pure \times Generational_Factor$	State + trajectory
N*	Narayama Index	$\sqrt{N_Base}$	Observable demographic core

8. WHY MULTIPLY? WHY THE SQUARE ROOT?

The multiplicative choice follows from the hypothesis of complementarity between stock and trajectory. An additive formulation would allow one dimension to fully compensate for deterioration in the other. For example, a society could have an extremely weak structure but a small recent improvement in fertility and still obtain an artificially high result. Multiplication reduces this possibility.

The square root, in turn, is a modeling choice intended to reduce the compression produced by multiplying the components and preserve an interpretable scale. These choices should not be presented as mathematical truths. They are specification hypotheses.

Validation of the index should assess: transparency; reproducibility; robustness; and additional information relative to conventional indicators. This methodological position is already explicitly formulated in the development of N*.

9. N* AND THE PENTAGONIA: TWO DIFFERENT FUNCTIONS

N* and the Pentagonia perform distinct functions within the Narayama framework. N* is the observable and measurable demographic core; the Pentagonia is the systemic architecture that explains the mechanisms that condition, amplify, or constrain its trajectory.

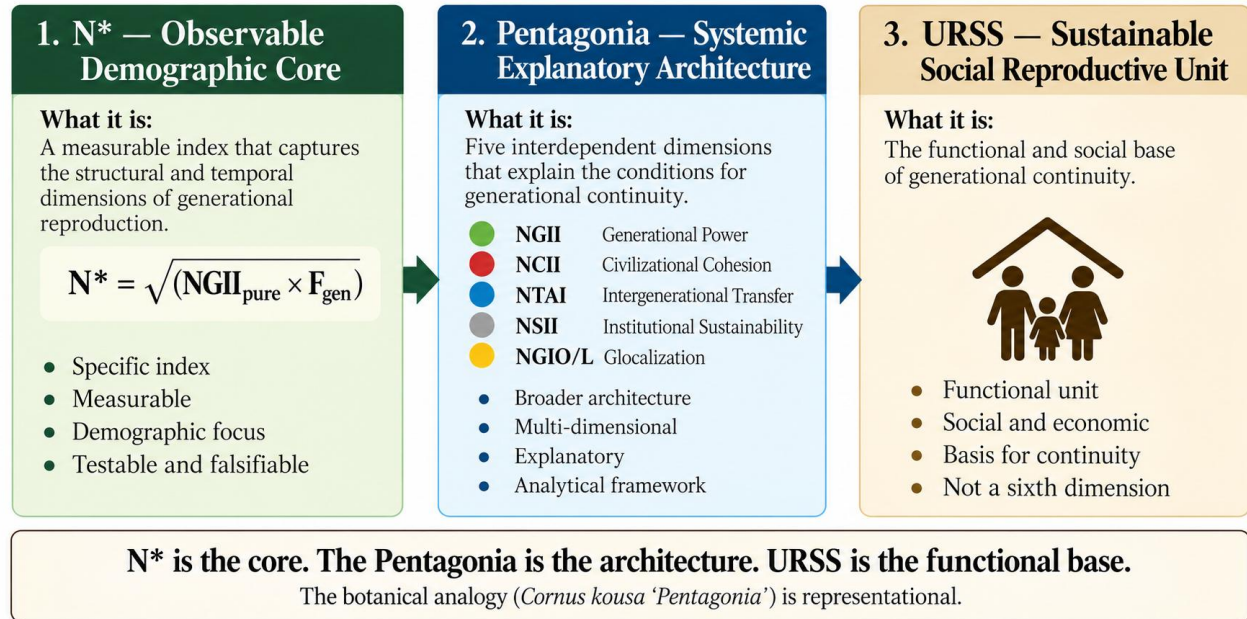


Figure 2 — N*, the Pentagonia, and the URSS: relationship between the demographic core, systemic architecture, and functional unit. N* is the observable demographic core; the Pentagonia is the explanatory architecture; the URSS is the functional base of generational continuity. The botanical analogy with *Cornus kousa* 'Pentagonia' is representational.

The Pentagonia consists of five dimensions: NGII — Generational Potency; NCII — Civilizational Cohesion; NTAI — Intergenerational Transfer; NSII — Institutional Sustainability; and NGIO/L — Glocalization. These dimensions are not additional components of the N* calculation. They function as explanatory layers for investigating why a given demographic trajectory occurs and which mechanisms may modify it.

The URSS occupies a distinct functional position. It is not a sixth dimension of the Pentagonia, but the social unit in which reproduction, care, formation, transmission, and integration of the next generation are articulated. Therefore, the URSS may be represented as the functional base or trunk of generational continuity, while N* remains the observable demographic core and the five dimensions of the Pentagonia form the explanatory architecture.

This distinction avoids conceptual overlap: N* answers the question “what is the condition and trajectory of generational renewal potential?”, while the Pentagonia helps investigate “which systemic mechanisms explain that condition and trajectory?”. The NTAI/Allocation Factor remains analytically separate from the central N* calculation, allowing future testing of whether the direction of transfers adds explanatory information to the demographic core.

Table 5 — Narayama Pentagonia: Five Pillars and Their Analytical Functions

Pillar	Code	Systemic function	Analytical question
Generational Potency	NGII	Capacity to generate and form new generations	Can society produce its next generation?
Civilizational Cohesion	NCII	Values, culture, and belonging	What connects generations?
Intergenerational Transfer	NTAI	Transmission of capital and knowledge	What is transmitted between generations?
Institutional Sustainability	NSII	Institutions and continuity	Can institutions support generational transition?
Glocalization	NGIO/L	Territory–global-system relationship	Where and at what scale does continuity occur?

The NTAI/Allocation Factor remains analytically separate from the central N^* calculation. This is important because it allows testing whether the direction of transfers adds explanatory information to the demographic core.

10. THE GENERATIONAL COST

From this point onward, the Pentagonia is used not as an abstract representation, but as a matrix for reading the costs produced by generational deterioration. Generational cost does not correspond to a single expenditure. It emerges from the need to reproduce, sustain, form, and integrate a new generation while the number of individuals among whom these functions can be distributed declines.

10.1 Reproductive Cost (NGII)

Reproductive cost lies at the core of the problem because it concerns a society's capacity to produce its next generation. NGII represents this structural potency. The URSS connects this demographic dimension to the material conditions of reproduction: income, housing, time, care, education, security, and future prospects.

10.2 Economic Cost (NGII + NSII)

Fewer children mean, decades later, fewer workers, consumers, entrepreneurs, and potential taxpayers. This transformation can reduce the scale of certain markets and alter investment, demand, and employment. A circuit may emerge in which economic contraction makes the formation of new URSSs more difficult, feeding back into low reproduction.

10.3 Fiscal Cost (NSII)

The reduction in younger generations progressively changes the relationship between those who contribute and those who depend on transfers, services, and collectively financed structures. The problem is not simply aging, but the asymmetry between accumulated obligations and the number of individuals who will have to sustain them in the future.

10.4 Institutional Cost (NSII + NTAI)

Institutions do not exist outside time. Social security, education, health care, labor markets, and systems of representation were built for particular population structures. When these structures change persistently, institutions must adapt. The question is whether the system can transmit rights and capacities without progressively concentrating obligations on smaller cohorts.

10.5 Territorial Cost (NGIO/L)

Generational contraction does not occur uniformly. When young families disappear from certain territories, schools close, services lose scale, commerce declines, and real-estate assets may lose economic function. Territory becomes part of the reproduction of decline itself.

10.6 Cultural Cost (NCII + NTAI)

Every generation receives knowledge, norms, practices, institutions, and references produced by previous generations and transmits part of them to the next. When generational continuity weakens, this transmission may suffer ruptures. The problem is not to prevent cultural transformation, but to understand under what conditions a society can transmit what it considers constitutive of its continuity when its successor generations become smaller.

The six costs are not independent problems. They represent different manifestations of the same rupture: fiscal → economic → reproductive → institutional → territorial → cultural. The Pentagonia makes this interdependence visible.

11. NARAYAMA HYPOTHESES

The paper carefully distinguishes observed evidence [E], hypothesis [H], and simulation [S], a convention already used in the larger project.

STATUS	MEANING
[E] Observed evidence	Demographic and economic phenomena observable in the data.
[H] Hypothesis	Relationships proposed by the Narayama framework and subject to testing.
[S] Simulation	Results derived from computational models and future scenarios.

H1 — Central Measurement Hypothesis

N* captures a dimension of generational contraction that is not captured in isolation by conventional demographic indicators.

H2 — Central Consequences Hypothesis

Persistent deterioration of generational potential is associated with measurable economic, fiscal, and institutional consequences.

Derived hypotheses (series agenda): H3 — demographic-economic feedback; H4 — migration as a buffer without restoration of endogenous reproduction; H5 — structural support for the URSS and reproductive trajectory; H6 — institutional and cultural continuity; H7 — systemic complementarity of the Pentagonia. Not all need to be demonstrated in this paper.

12. EMPIRICAL STRATEGY

The paper does not need to pretend that all hypotheses have already been demonstrated. Its strength lies in presenting a testable architecture.

The proposal works with a panel of 28 countries (1990–2024), demographic indicators, TFR, age structure, births, deaths, NTA where available, and complementary economic and institutional variables. The larger project envisages a fixed-effects panel with interaction terms, in addition to OLG simulations.

The first empirical question should be: does N* provide additional information relative to TFR, median age, dependency ratio, and population growth? This is the most important question for validating the index.

Other planned tests: retrospective backtesting; robustness (20/25/30-year horizons, different age groups, alternative transformations, inclusion/exclusion of migration); and a falsifiability protocol capable of confirming, weakening, or refuting the specifications.

13. LIMITATIONS

The paper recognizes from the outset:

1. The 25-year horizon is an operational approximation, not a biological constant.
2. Cohort definitions may produce different results under alternative specifications.
3. Separating endogenous reproduction from migratory effects is methodologically difficult.
4. Data quality and comparability vary across countries.
5. Association does not imply causality.
6. N^* does not measure all civilizational sustainability; the Pentagonia is broader.
7. The choice of the square root is a modeling decision and should be confronted with alternatives.
8. NTAI is not directly incorporated into the N^* core.

This is important because the paper gains credibility precisely by not presenting the model as finished.

14. DISCUSSION

14.1 Longevity Is Not the Problem

Longevity is one of civilization's great achievements. The problem appears when the capacity to prolong lives is no longer accompanied by an equivalent capacity to produce and sustain subsequent generations. The problem is not living longer. The analytical problem is living longer while the capacity to produce the next generation persistently declines.

14.2 Technology Does Not Replace Social Reproduction

Automation can replace certain labor functions. It does not fully replace birth, formation, socialization, family transmission, and intergenerational continuity. Technology can replace certain economic functions performed by human beings; it does not replace the biological and social reproduction required to produce future generations themselves.

14.3 Older Generations Are Also Transmitting Heritage

Older generations are not merely recipients of transfers from younger generations. They also constitute one of the principal bases for transmitting the knowledge accumulated by society. The generational question has two directions.

14.4 The Demographic Inversion of Knowledge

The greater the longevity, the larger the accumulated stock of experience potentially available for transmission. The smaller the successor generation, the fewer individuals are available to receive, transform, and retransmit this stock. This turns aging into not only a fiscal but also an epistemic and civilizational issue.

14.5 Immigration Is Not Reproduction

Immigration can compensate for population. It is not equivalent to the recovery of endogenous fertility. These are different mechanisms and should be measured separately.

14.6 Territory Matters

The same national fertility rate can produce radically different consequences in dense cities, metropolitan areas, small cities, and depopulating regions. For this reason, NGIO/L is not an ornamental element of the Pentagonia. It is the territorial dimension that conditions the environment in which the URSS exists.

15. CONCLUSION

The Narayama Index does not arise from an attempt to create yet another demographic indicator. It arises from a question.

Contemporary societies have developed extraordinary capacity to prolong life, preserve assets, increase productivity, and use technology to substitute for human labor. But these same societies must answer a question that none of these achievements eliminates:

who will receive, sustain, and transform what we are building?

Declining fertility is not simply a statistical change. It modifies the future structure of the labor force, demand, taxation, institutions, territories, and mechanisms of intergenerational transmission.

It is in this context that the URSS becomes a relevant economic unit — with the traditional form of natural providers (father, mother, grandfather, and grandmother) as its mater and majority cell, while recognizing other forms insofar as they effectively assume the costs of forming the next generation, but without placing them on the same plane.

It is in this context that NGII seeks to represent generational potency; the Generational Factor seeks to represent the direction of the trajectory; and N^* emerges from the combination of these two elements.

The Narayama Pentagonia expands this reading. It proposes that intergenerational sustainability cannot be reduced to the number of births. It involves generational potency, civilizational cohesion, intergenerational transfer, institutional sustainability, and glocalization.

N^* is the core. The Pentagonia is the architecture. And generational cost is what appears when the core deteriorates and its consequences cross the rest of the system.

The contemporary demographic challenge is not simply to manage an older population. It is to understand a society in which the longevity of existing generations increases at the same time that the number of individuals destined to succeed them decreases.

The purpose of this paper is not to claim that a particular society should grow, shrink, receive immigrants, automate its economy, or adopt a particular family policy. It is prior to that. It is to create a way to ask, measure, and test:

is a society still producing the economic, social, and institutional conditions necessary for a generation capable of succeeding it to exist?

If the answer is negative, the problem does not begin when the population disappears. It begins much earlier. It begins when society ceases to perceive that those who have not yet been born are also part of its economic equation.

Testing this hypothesis — in a robust, comparable, and falsifiable way across different societies and periods — is the necessary next step.

And perhaps this is Narayama's fundamental inversion:

we are no longer a society that must decide which elderly people it can sustain. We are societies that must decide whether we are still building the conditions for those who will have to sustain, transform, and surpass the world they inherit to exist.

A civilization reproduces itself not only when new individuals are born. It reproduces itself when one generation is able to form the next, transmit knowledge and assets to it, integrate it into its institutions, and later receive from it the continuity of that same chain. This is not moralism. It is precisely the ethical-civilizational postulate that gives meaning to the URSS within Narayama.

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